911 Emergency Call Volume Prediction using Machine Learning

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Abstract

This project develops a machine learning model to **predict hourly 911 call volume** across EMS, Fire, and Traffic departments. Using time-based features and historical data, we identify patterns in emergency demand to **improve resource planning** and short-term forecasting.

Objective

- Predict hourly emergency call volume
- Identify time-based demand patterns
- Compare performance of ML models
- Support better resource allocation

Challenges

- Highly **skewed distribution** with rare peak-demand spikes
- **Weak linear relationships →** limits effectiveness of simple models
- Strong **temporal dependency** (data cannot be randomly shuffled)
- Difficulty in accurately predicting **high-demand periods**
- Risk of overfitting due to engineered interaction features

Framework

Data Preparation

- Clean data
- Extract time features

Feature Engineering

- Hour, day, month
- Weekend flag
- Dept encoding
- Interaction terms

Data Transformation

- Aggregate → hourly data
- Handle zero-call intervals
- Scale & encode

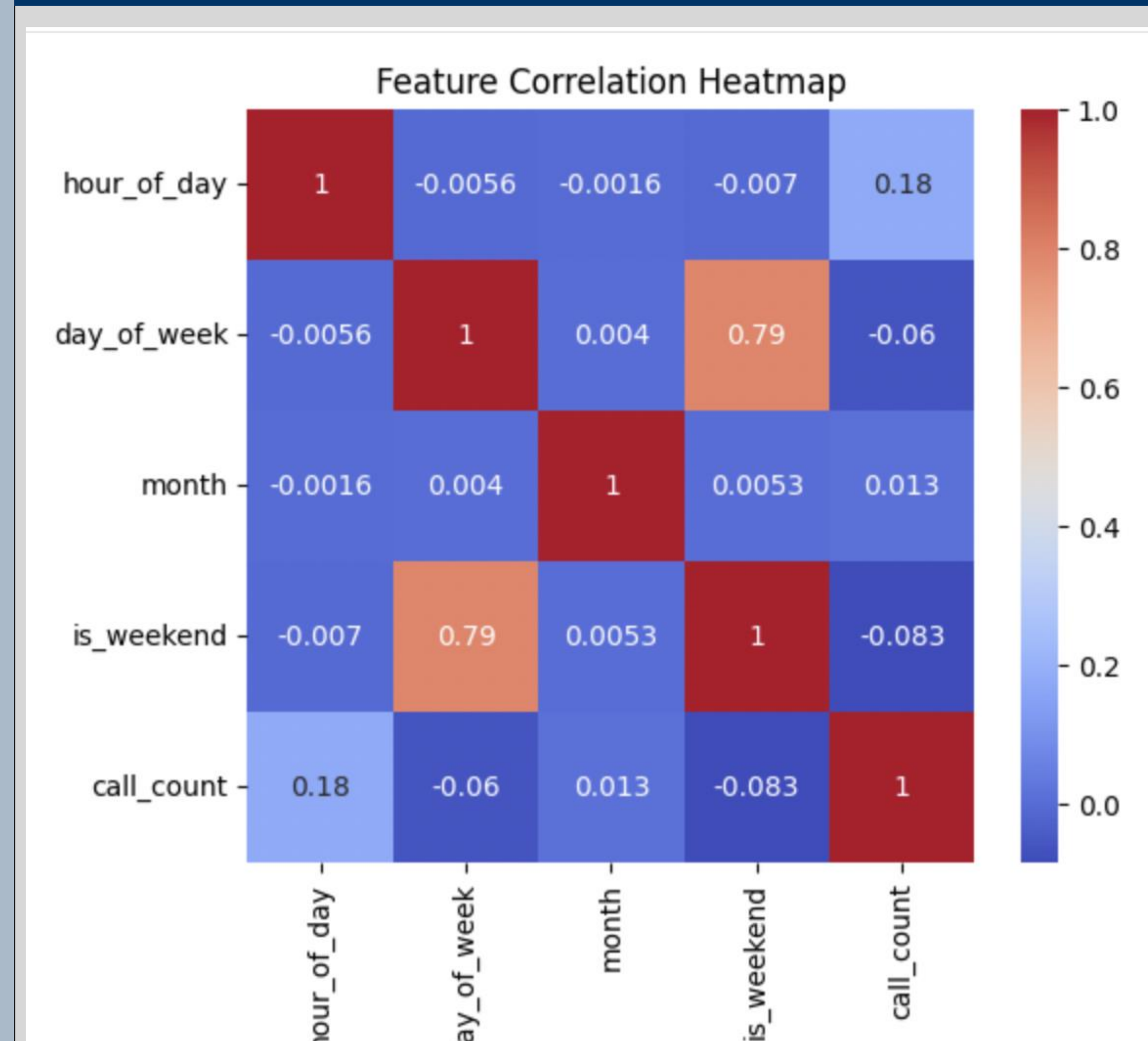
Model Training

- Linear, RF, GB
- Time-based split

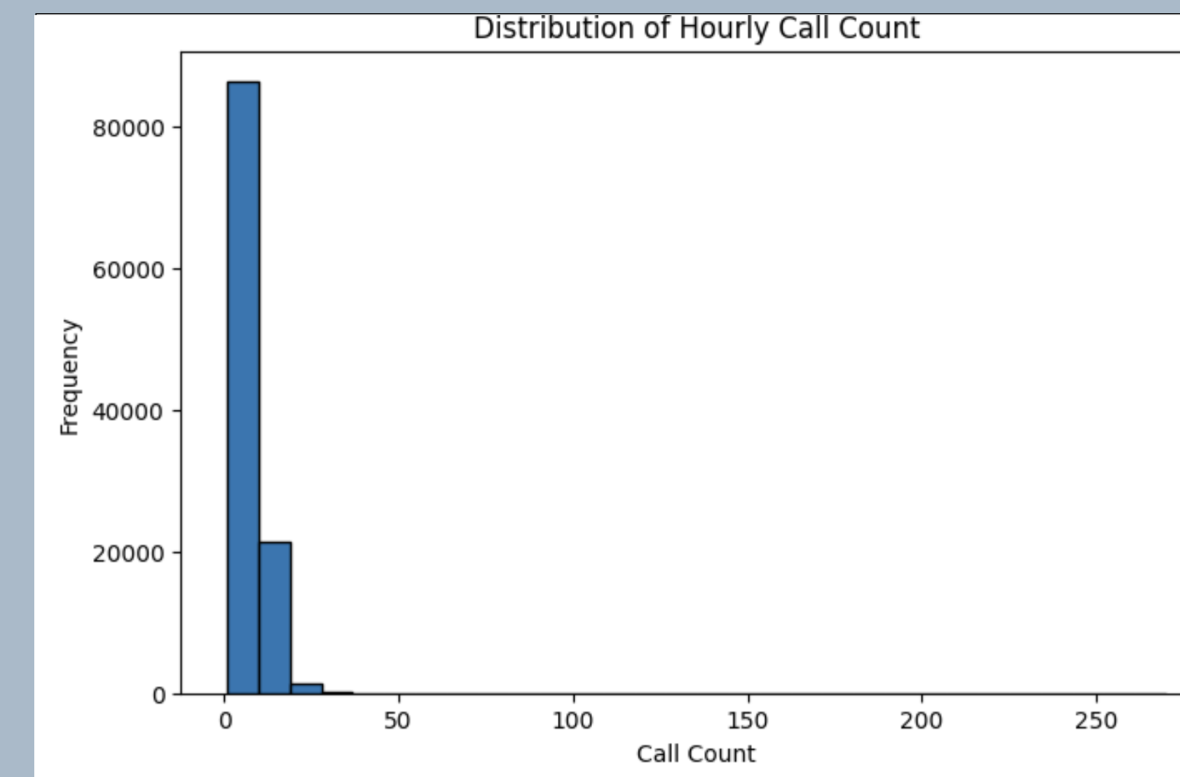
Evaluation & Selection

- MAE, RMSE, R^2
- Select best model

Examples



Time has strongest influence



Data is right-skewed

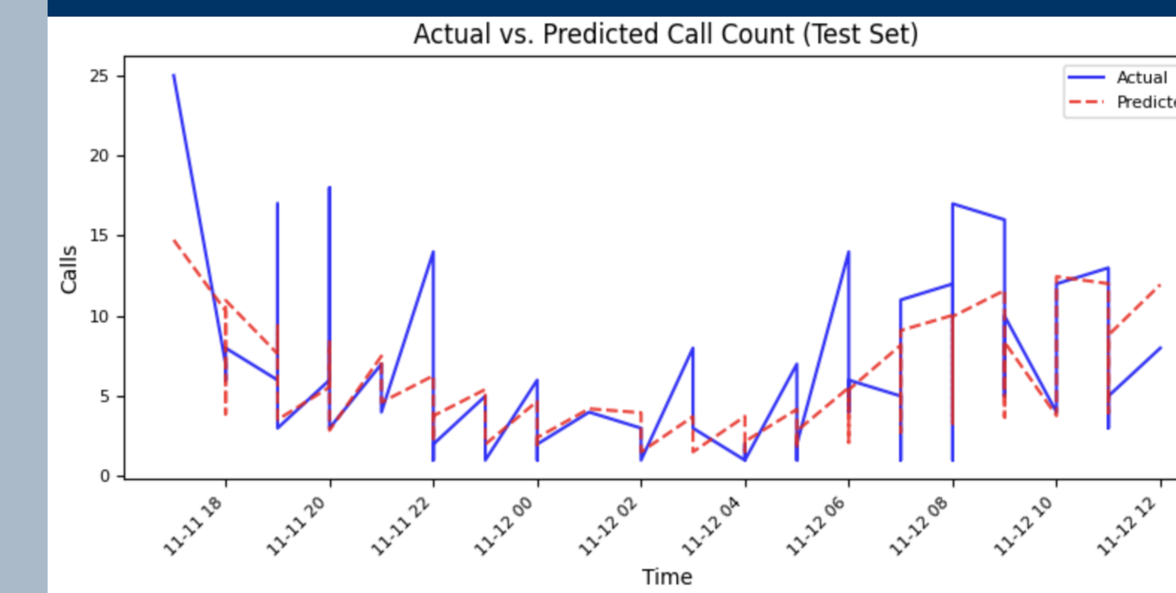
Evaluations

	Model	MAE	RMSE	R^2
0	Linear Regression	2.95	4.08	0.28
1	Random Forest	2.13	3.27	0.56
2	Gradient Boosting	2.13	3.20	0.56

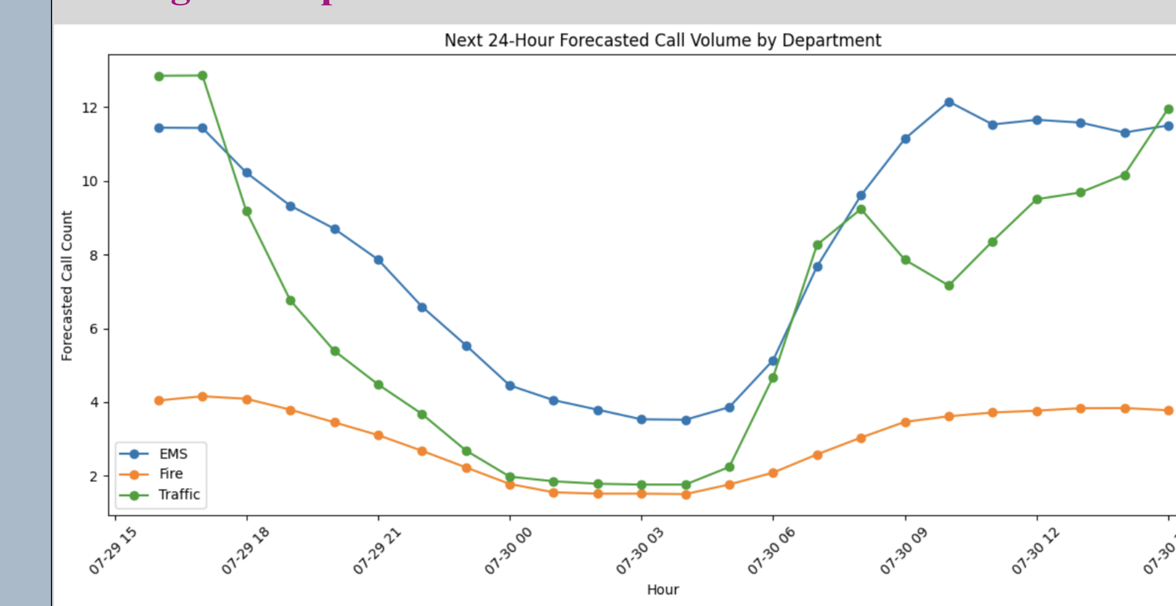
- Linear Regression underperforms due to inability to capture non-linear patterns
- Tree-based models better handle skewed data and complex relationships
- Gradient Boosting delivers the most stable and accurate results

Gradient Boosting provides the most reliable predictions with the lowest error across datasets.

Outcomes



Predicted values closely track actual call volume, indicating strong model performance.



Distinct demand patterns are observed across departments, with predictable daily trends.

- Predictions **closely match actual call patterns**
- Captures clear **time-based demand trends**
- Reliable for **short-term forecasting**

Key Takeaways

- Emergency call volume is strongly driven by **time-based patterns**
- Linear Regression is insufficient for capturing complex relationships
- **Tree-based models significantly improve prediction accuracy**
- Gradient Boosting provides the **most reliable and consistent results**

Impact:
Improves planning and response efficiency.

References

1. Montgomery County 911 Dataset (Kaggle)
2. Scikit-learn Documentation (Machine Learning Models)
3. Course Materials , Seneca Polytechnic